

AGRIVOLTAIC SYSTEM DESIGN: BALANCING SPATIAL CONSTRAINTS, CROP
YIELDS, AND ENERGY GENERATION

A Thesis Proposal

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ABSTRACT

Agrivoltaic System Design: Balancing Spatial Constraints, Crop Yields, And Energy Generation

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As utility-scale solar development accelerates, escalating land-use conflicts threaten critical agricultural sectors. Agrivoltaics creates a symbiotic relationship that benefits communities, crops, and farmers by integrating solar panels into agricultural systems. This infrastructure provides shade, reduces water evaporation, and shields plants from extreme heat while generating electricity. Economically, co-location provides farmers with year-round income, increases crop yields, preserves water, and enhances electrical grid resilience. However, current systems are largely limited to hand-harvested crops, as panel supports obstruct the heavy, mechanized equipment essential to large-scale staple crops such as cotton and grain sorghum. Furthermore, existing funded programs primarily benefit northern states, leaving a significant need for fundamental research in the southern United States. This research tackles the fundamental question: How can agrivoltaic systems be designed to minimize disruption to mechanized agricultural workflows while maximizing dual-use land profitability? To answer this, the study pursues two primary objectives. Objective 1 aims to investigate the physical impacts of solar infrastructure on mechanized farming. This will be completed by conducting swept-volume analyses of agricultural machinery to define strict spatial clearance requirements, paired with shading simulations to quantify localized crop yield impacts. Objective 2 seeks to develop a stochastic decision-making framework to identify the optimal spatial layout. This will be completed by first deploying a TOPSIS (Technique for Order of Preference by Similarity to Ideal Solution) matrix to mathematically standardize the competing mechanical, biological, and economic tradeoffs. These parameters will then be integrated into a Two-Stage Stochastic Mixed-

Integer Linear Programming (MILP) algorithm. By incorporating subseasonal AI weather forecasts to account for environmental volatility, the algorithm dynamically balances energy generation and crop yields to maximize the site's 25-year Net Present Value (NPV). Ultimately, this research yields a mathematically validated decision-making tool that facilitates renewable energy expansion without compromising the operational efficiency of mechanized agriculture.

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CHAPTER I: INTRODUCTION

The global transition to a decarbonized energy system has positioned solar photovoltaic (PV) technologies at the forefront of climate change mitigation strategies [1]. However, the rapid and extensive deployment of large-scale, ground-mounted solar power plants requires vast amounts of land. This spatial footprint has ignited a severe land-use conflict, as renewable energy development increasingly competes directly with fertile agricultural land, fueling the "energy versus food" debate and raising concerns over long-term food security [2]. To reconcile this competition at the Water-Energy-Food nexus, agrivoltaic systems (AVS)—the simultaneous, dual-use co-location of solar energy generation and agricultural production on the same parcel of land—have emerged as a highly promising solution [3], [4]. By sharing available sunlight, agrivoltaics can significantly improve the overall land efficiency, a synergistic benefit typically quantified by the Land Equivalent Ratio (LER) [5], [6]. Despite these mutual benefits, integrating solar infrastructure into active agricultural environments introduces complex spatial, agronomic, and ecological challenges. The physical presence of solar modules fundamentally alters the crop microclimate. Most notably, the panels reduce the amount of Photosynthetically Active Radiation (PAR) reaching the underlying canopy, which can inhibit the growth, biomass accumulation, and yield of shade-intolerant crops if not carefully managed [5], [7]. Furthermore, unoptimized and rigid permanent structures can physically interfere with day-to-day farming operations, restricting the maneuverability of heavy agricultural machinery and potentially degrading soil health through compaction or altered hydrological runoff. Because of these inherent trade-offs, designing an effective agrivoltaic system is not merely about placing solar panels in a field; it requires a multidimensional optimization of spatial constraints. The design must carefully balance key architectural parameters—such as Ground Coverage Ratio (GCR), inter-row spacing (pitch), panel

clearance height, and seasonal tilt angles—against the physiological light requirements of the specific crops being cultivated. While traditional solar farms are engineered strictly to maximize electricity output, a sustainable agrivoltaic design must prioritize agricultural viability. Achieving this requires moving beyond static panel configurations to explore dynamic solutions, such as horizontal single-axis trackers (HSAT) or dual-axis trackers, which can adapt to real-time crop light thresholds and ensure a more homogeneous distribution of shade and rainfall. This thesis explores the critical intersection of system engineering, agronomy, and spatial planning to establish a comprehensive design framework for agrivoltaic systems. By moving beyond isolated studies of either energy maximization or crop survival, this research investigates how advanced light-environment simulations and structural layout optimization can be harmonized. Ultimately, this study demonstrates how strategically balancing spatial constraints can mitigate agricultural interference, maximize the dual-yield of crops and clean energy, and ensure that the transition to renewable technologies enhances rather than compromises global agricultural resilience.

1.1 Background and Problem Statement

As solar energy infrastructure scales to meet global energy demands, the physical footprint of solar farms increasingly encroaches on arable land. While integrating PV technology directly into agricultural production areas (agrivoltaics) mitigates this land-use conflict, the physical presence of solar structures introduces complex operational penalties. Different structural configurations—such as fixed-tilt, single-axis tracking, dual-axis tracking, and elevated systems—impose varying physical constraints on standard farming activities, as shown in figures 1-4.



Figure 1: Single-Axis Tracking System [8]

For instance, single-axis tracking systems (Figure 1) follow the sun's trajectory throughout the day, creating dynamic shading patterns; however, their long, continuous drive-rows can severely restrict the cross-field maneuverability of heavy tractors.



Figure 2: Dual-Axis Tracking [9]

Dual-axis trackers (Figure 2) maximize energy capture through a wide, multi-directional sweep, but their massive centralized support pillars require significant spatial buffers, reducing the total workable land area.



Figure 3: Elevated Tilted System [10]

Elevated fixed-tilt systems (Figure 3) are explicitly designed to provide permanent overhead clearance for crops, yet the sheer density of their foundational supports can still obstruct wide-boom sprayers and large combine harvesters.



Figure 4: Vertical-Bifacial [11]

Lastly, vertical bifacial arrays (Figure 4) are mounted at a 90° angle to preserve wide corridors for farming; however, their continuous, ground-level structural rows create rigid, linear boundaries that prevent lateral cross-field movement and dictate fixed operational routes.

These configurations can obstruct the kinematic pathways of tractors and harvesters, reduce workable land area, alter crop microclimates through persistent shading, and introduce operational downtime during severe weather events. To address these challenges, this research is guided by the following core question and objectives.

Research Question: How can the impact of a different agrivoltaic system on agricultural activities be minimized?

Research Objective-1: Investigating the agrivoltaic impact on agricultural activities in differing crops and agrivoltaic systems.

Research Objective -2: Develop a decision-making framework to minimize the impact and identify the optimal placement of agrivoltaics across different agricultural activities.

1.2 Schedule of Thesis

Target Graduation: December 2026. The core data synthesis and methodological structuring are designated for the Spring 2026 academic term.

Spring 2026 (Jan – Mar): Conduct systematic literature review on existing agrivoltaic systems and compile datasets of crop light requirements alongside standard farm machinery dimensions (tractor widths/turning radii).

Spring 2026 (Apr – May): Complete CAD-based kinematic clearance testing (SolidWorks) and shading profile simulations (NREL SAM). Establish the structural constraints for differing AVS configurations.

Summer 2026 (Jun – Aug): Extract subseasonal-to-seasonal weather forecasting ensembles (AERIS AI model) at Argonne National Laboratory to define probabilistic wind constraints, operational downtime, and irradiance variances.

Fall 2026 (Sep – Oct): Develop and test the decision-making framework via a Two-Stage Stochastic MILP solver in Python, integrating mechanical constraints and weather probabilities.

Fall 2026 (Nov – Dec): Finalize thesis manuscript, complete committee review, and conduct Master's Thesis Defense.

CHAPTER II: LITERATURE REVIEW

2.1 Contextual Landscape and the Global Energy-Land Nexus

The global energy landscape is currently undergoing a systemic transition, pivoting away from fossil-fuel dominance toward decentralized, cleaner energy systems to mitigate climate change.[4]. Solar energy has established itself as the primary catalyst for this shift; however, the rapid expansion of utility-scale ground-mounted solar infrastructure frequently creates severe land-use conflicts, directly competing with agricultural production [2]. This issue is exacerbated in regions where population growth and arable land depletion stretch the delicate balance of the water-energy-food nexus [4].

Agrivoltaic systems (AVS) have emerged as a necessary technological compromise to overcome local zoning barriers while generating multifaceted socioeconomic benefits. Case studies demonstrate that AVS can effectively navigate strict local land-use policies by preserving agricultural functionality [12]. The viability of these dual-use systems is typically measured using standardized metrics, such as the Land Equivalent Ratio (LER) and policy constraints like the Land Area Occupation Ratio (LAOR) [3], which quantify the combined land-use efficiency of co-locating crops and solar panels. However, as the literature reveals, achieving a high LER is heavily dependent on the system's physical design, which currently imposes severe limitations on broad-acre farming.

Beyond pure metrics, the socioeconomic and psychological dimensions of solar deployment present significant hurdles. Rural communities frequently resist large-scale solar projects due to the perceived industrialization of rural landscapes and the loss of agricultural heritage [12]. Agrivoltaics offers a pathway to mitigate this public dissent by maintaining the aesthetic and functional characteristics of working farms. Projects like Jack's Solar Garden

illustrate that when local farmers retain ownership and agricultural output remains robust, community acceptance rises, paving the way for educational outreach and local industry integration [12]. Consequently, any proposed AVS design must not only satisfy energy and crop metrics but also remain culturally and operationally acceptable to the farmers who manage the land.

2.2 Hardware Engineering and the Rigidity Problem

The physical implementation of conventional agrivoltaics relies on permanent hardware configurations—primarily static tilted, vertical bifacial, and horizontal single-axis tracking systems. [1], [5]. While vertical bifacial farms offer wider corridors for farming by utilizing east/west orientations to capture reflected albedo radiation [1]. The vast majority of current AVS infrastructure uses deep-driven steel piles and rigid overhead canopies.

This presents a critical gap in the literature and industry practice: traditional rigid project management and construction approaches fail to account for the dynamic spatial needs of modern agriculture. The fixed nature of these structures inherently limits their application to shade-tolerant, hand-harvested horticultural crops (e.g., lettuce, berries) [13]. They are fundamentally incompatible with large-scale mechanized agriculture (e.g., combine harvesters for wheat, corn, and cotton), which forms the backbone of global food security.

Furthermore, the engineering of these permanent structures is heavily dictated by structural load requirements. Overhead canopies must be engineered to withstand severe wind and snow loads, necessitating massive steel foundations and complex mounting racks, which increase capital expenditures (CapEx). Recently, portable and modular agrivoltaic hardware has been proposed to increase system adaptability, enabling panels to be moved or reconfigured to accommodate

seasonal access by machinery [5]. However, the structural optimization of these highly adaptable systems remains under-researched.

2.3 Agricultural Integration and Interference

Introducing solar infrastructure into agricultural spaces inherently alters the crop microclimate. PV modules serve as an abiotic buffer that reduces soil and air temperature extremes, increases relative humidity, and significantly lowers water evaporation rates [14]. This dynamic can be highly beneficial in drought-prone environments, as seen in tomato production under AVS, where lower soil water gradients led to increased water use efficiency [15].

Conversely, panels block Photosynthetically Active Radiation (PAR), which heavily impacts shade-intolerant crops. The extent of this impact is measured by the Daily Light Integral (DLI) and the spatial uniformity of the light reaching the canopy. Crop-specific physiological responses—such as delayed acclimation in rice paddies [16] and reduced dry matter accumulation in soybeans [7]—are highly sensitive to the spatial heterogeneity of this shade. Fixed-tilt panels often cast dense, persistent shadow bands that cause uneven crop growth, whereas tracking systems or strategically spaced portable arrays can distribute light more evenly, preventing localized yield failures [5], [13]

Beyond shading, the most profound impact of AVS is human and mechanical interference. The permanent spacing of structural supports disrupts day-to-day agronomic practices, forcing farmers to alter irrigation schedules, abandon standard crop rotations, and use smaller, less efficient machinery [5], [14]. Minimizing this mechanical interference through optimized spatial systems is not just a secondary benefit; it is a primary requirement for the widespread adoption of agrivoltaics.

2.4 Environmental, Hydrological, and Ecological Impact

While AVS offers climate-mitigation benefits, permanent installations incur localized environmental costs, strengthening the case for adaptable alternatives. The "roof effect" generated by continuous panel arrays alters natural hydrological connectivity. This concentration of surface runoff exacerbates soil erosion dynamics, particularly in hilly or sloped environments where rainfall is channeled abruptly off panel edges [17].

Furthermore, the construction of permanent foundational structures and the navigation of heavy installation machinery lead to long-term physical soil compaction and degradation [18]. The initial construction phase involves extensive trenching for electrical conduits and the pounding of deep steel pilings, which disrupt delicate soil horizons, sever root networks, and damage the soil microbiome (Figure 5).



Figure 5: Agrivoltaics trenching for wire installations [19]

From a broader perspective, Life Cycle Assessments (LCA) on large-scale solar plants reveal a damage category distribution heavily affecting ecosystem quality and land resources [20]. Portable agrivoltaic systems offer a theoretical pathway to reduce permanent soil degradation. By

utilizing above-ground ballasts or shallow, temporary anchoring systems, and by allowing the modules to be relocated to rest degraded land, non-permanent systems enable ecological restoration and allow native plant communities to recover naturally [18].

2.5 The Algorithmic Gap in Decision-Making Frameworks

Designing an optimal AVS requires balancing competing priorities: maximizing energy output, maintaining agricultural viability, and minimizing logistical interference [6]. Current trade-off analyses indicate that while adjusting module tilt and Ground Coverage Ratio (GCR) slightly reduces peak energy yield, it drastically improves the light environment for the crops below [21]. Currently, advanced digital replicas and 3D ray-tracing simulations are utilized to map high-resolution irradiance profiles [22]. These models use Monte Carlo backward ray-tracing algorithms to simulate light reflection and scattering, generating massive crop-level PAR datasets. These datasets are then fed into crop growth simulators like GECROS or SIMPLE to predict biomass accumulation [21], [22].

Despite the sophistication of these environmental models, current mathematical programming frameworks are fundamentally incomplete. Existing optimization algorithms focus almost exclusively on light-sharing and energy yield [6], [22], entirely omitting spatial penalties for machinery swept-volume or dynamic geospatial restrictions [23]. Integrating mechanical spatial constraints into genomic algorithms creates a highly complex, non-linear optimization problem. To truly optimize agrivoltaic systems for broad-acre farming, a novel algorithm must be developed that introduces human activity, kinematic machinery clearance, and dynamic spatial reallocation as primary mathematical constraints within the cost function.

CHAPTER III: METHODOLOGY

3.1 Study Area and System Parameters

To address the core research question regarding machinery interference and economic optimization in hot, water-scarce environments, this study localizes its framework within the semi-arid climate of South Texas, specifically modeling locations such as El Paso and Mission. This region is characterized by extreme climate stress (+2–4°C projected warming) and severe water scarcity, necessitating a 30–50% reduction in agricultural irrigation, making it highly suitable for climate-resilient land-use frameworks [24]. The simulations use a sandy loam soil profile (120–140 mm m⁻¹ available water capacity), representative of typical southwestern U.S. irrigated agriculture.

To evaluate the spatial, biological, and energy impacts of mechanized farming, four distinct agrivoltaic (AV) structural configurations are assessed. These configurations vary significantly in their tracking mechanisms, tilt and rotation limits, ground coverage ratios (GCR), and machinery clearance heights [22], [23]. GCR represents the proportion of total land area directly covered by the solar modules, serving as a critical geometric parameter that dictates both the density of the energy array and the amount of ambient sunlight penetrating the crop canopy [25].

The first configuration, a Vertical Bifacial system, relies on fixed mounts at a 90° vertical angle, featuring a minimal GCR of 0.10 and a low 0.8-meter ground clearance. The second setup, a Single-Axis tracker, rotates along one axis with a ±55° limit; it occupies the highest land area with a 0.35 GCR and provides a moderate 2.5-meter clearance. The third configuration is an Elevated Fixed array, permanently tilted 15° facing south, offering a 0.25 GCR and a maximum constant clearance of 5.0 meters to easily accommodate tall farming equipment. Finally, the Dual-Axis system utilizes full, two-axis solar tracking to maximize energy yield, maintaining a 0.25

GCR while operating with variable clearance heights depending on its dynamic positioning. Broad-acre staple crops, specifically cotton and corn, are selected as the biological controls based on regional economic data. Table 1 summarizes the information related to four configurations.

Table 1: Photovoltaic Configuration Specifications for Comparative Analysis

Configuration	Tracking	Tilt/Rotation	GCR	Clearance
Vertical Bifacial	Fixed	90° vertical	0.10	0.8 m
Single-Axis	1-axis	0° ($\pm 55^\circ$ limit)	0.35	2.5 m
Elevated Fixed	Fixed	15° south	0.25	5.0 m
Dual-Axis	2-axis	Full tracking	0.25	Variable

3.2 Research Objective 1: Investigating the Agrivoltaic Impact on Agricultural Activities in Differing Crops and Agrivoltaic Systems

3.2.1 Task 1.1: Defining Agricultural Impacts and Human Interference

Because a primary barrier to agrivoltaic adoption in staple-crop farming is the restriction of heavy machinery, the first phase of this methodology maps out the exact physical disruptions caused by solar infrastructure. The current literature highlights that conventional rigid mounting structures often inhibit the use of heavy agricultural machinery [5]. Advanced kinematic swept-volume analyses will be conducted using SolidWorks. This modeling will simulate the turning radii, boom widths, and standard operational routes of required agricultural machinery (e.g., tractors and combine harvesters). By identifying the most severe physical bottlenecks, the study will lock in hard physical clearance requirements that act as non-negotiable geometric constraints for the final system layout [3].

3.2.2 Task 1.2: Ranking the Highest Impacts

Once all physical and logistical impacts are investigated via the 3D kinematic models, this task ranks them to find the absolute worst physical bottlenecks. For example, the study will identify

how a single-axis tracker stowed at a specific angle might completely block a headland turn for a combine harvester, or how structural pillar spacing dictates maximum implement widths [5].

By categorizing and identifying the most severe physical impacts, the methodology isolates exactly which structural barriers the final optimization layout must prioritize minimizing to maintain broad-acre agricultural viability.

3.2.3 Task 1.3: Maximizing Yields through Environmental Simulation

To directly investigate the biological impact of differing agrivoltaic systems on various staple crops, a coupled digital-replica modeling approach is utilized.

Energy Generation Modeling: The National Renewable Energy Laboratory’s System Advisor Model (NREL SAM v2025.4.16) is utilized to generate high-resolution 3D shading profiles and hourly energy production data [21]. The simulations assume identical 237.6 kW DC installations equipped with Mission Solar MSN10-550HT4G bifacial modules across all configurations [24].

Agronomic Modeling: Coupled with the Decision Support System for Agrotechnology Transfer (DSSAT v4.8.2.0) crop model, this task quantifies yield impact factors based on localized crop light and water requirements, directly modeling evapotranspiration reduction and biomass accumulation [7]. The CROPGRO-Cotton and CERES-Maize models simulate broad-acre staple crop responses to baseline (0%), moderate (30%), and intensive (50%) shading scenarios across a six-year climate period (2018-2023). This identifies which panel configurations cause the least biological harm to the crops below [14], [15], [24].

3.2.4 Task 1.4: Panel Maintenance Integration

Finally, this step investigates the operational footprint of the solar array itself. It tracks how the humans and trucks needed to clean and fix the panels interfere with farm space. Factoring in

this maintenance space ensures the final framework minimizes physical traffic jams between farm workers and solar technicians [5].

3.3 Research Objective 2: Develop a Decision-Making Framework to Minimize Impact and Identify Optimal Placement

3.3.1 Task 2.1: Developing the Scope for Optimization

This task establishes the mathematical boundaries for the decision-making framework. It sets up the decision variables and constraints, deliberately incorporating subseasonal AERIS AI weather data from the Argonne National Laboratory to handle unpredictable climate conditions. Defining this scope translates the real-world goal of "minimizing impact" into concrete mathematical formulas that a computer algorithm can process. Utilizing a digital-replica framework ensures the model responds dynamically to real-world environmental shifts rather than static historical averages [22].

3.3.2 Task 2.2: Finding the Trade-off Between Energy and Crop Production

Balancing the competing goals of an agrivoltaic system requires standardizing complex variables across energy, agriculture, and mechanical engineering. This task builds the core logic for the decision-making framework by determining the financial formula that weighs the revenue of generating solar power against the cost of lost crops or lost farming space. The Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) framework will be employed [26].

Table 2 presents the weighted criteria used to rank the performance of the modeled configurations under the "ideal" and "negative-ideal" scenarios [27]. Economic and agronomic outcomes are heavily prioritized, with Total Net Present Value (NPV) carrying the highest weight (30%), where single-axis tracking scores a perfect 10/10 compared to a low 3/10 for vertical bifacial systems. Crop Preservation follows closely at 25% weight, with single-axis again leading

(10/10) over dual-axis (8/10) and vertical bifacial (6/10) arrays. Capital Efficiency (20%) strongly favors dual-axis systems (10/10) over single-axis systems (4/10). The remaining 25% of the decision weight is distributed among operational factors: Machinery Access (10%), Operations and Maintenance (O&M) Simplicity (10%)—where vertical bifacial systems excel (10/10) while complex dual-axis trackers struggle (3/10)—and Grid Integration (5%).

Table 2: Multi-Criteria Comparison Framework Weights and Baseline Scoring

Criterion	Weight	Vertical Bifacial	Single-Axis	Dual-Axis
Total NPV	30%	3/10	10/10	8/10
Crop Preservation	25%	6/10	10/10	8/10
Capital Efficiency	20%	8/10	4/10	10/10
Machinery Access	10%	7/10	6/10	7/10
O&M Simplicity	10%	10/10	9/10	3/10
Grid Integration	5%	8/10	5/10	6/10

3.3.3 Task 2.3: Developing the Framework and Algorithm

To definitively identify the optimal layout, a Two-Stage Stochastic Mixed-Integer Linear Programming (MILP) solver will be constructed in RStudio. The MILP solver will ingest the constraints derived from the SolidWorks kinematic analysis, the environmental profiles from NREL SAM/DSSAT, and the weighted criteria from the TOPSIS framework. The financial parameters fed into the algorithm include a \$0.12/kWh Power Purchase Agreement (PPA) rate with Time-of-Day (TOD) multipliers (3.5x summer peak, 2.0x winter peak), a 30% Federal ITC, and a \$56,250 AEP SMART Source Rebate. The solver will conduct rigorous sensitivity analyses on market pricing and weather variables [27]. The final output of this algorithm provides the specific, mathematically proven layout that minimizes agricultural disruption.

3.4 Project Timeline

The research will be executed according to the following schedule to ensure systematic data integration and model building:

- **April 2026:** Execute SolidWorks kinematic swept-volume analyses and lock in hard physical clearance constraints.
- **May 2026:** Generate 3D shading profiles utilizing NREL SAM and calculate yield impact factors based on localized crop requirements.
- **June – July 2026:** Develop the financial objective function, extract and integrate subseasonal AERIS AI weather data from Argonne National Laboratory, and define parameters for the case study.
- **August – September 2026:** Construct the Two-Stage Stochastic MILP solver in RStudio, run the South Texas case study, and conduct rigorous sensitivity analyses on market pricing and weather variables.
- **October 2026:** Format software code, organize CAD diagrams, compile final data, and submit the final Master's dissertation manuscript.
- **Early November 2026:** Conduct thesis defense and gather feedback from the committee.
- **Late November – December 2026:** Implement final revisions, secure formatting approval from the graduate school, and finalize all graduation requirements.

CHAPTER IV: EXPECTED OUTCOMES

4.1 Geometric and Kinematic Guidelines for Mechanized Agriculture

By successfully executing Research Objective 1, this study will produce a definitive, quantified set of geometric clearance guidelines for agrivoltaic infrastructure. By utilizing the 3D SolidWorks kinematic simulations, the research will isolate exact structural bottlenecks—such as structural pillar spacing constraints and tracker stow-angle obstructions. These standardized parameters will provide solar developers with the exact spatial specifications required to safely maneuver large-scale machinery, effectively eliminating a primary logistical barrier to dual-use farming.

4.2 Quantification of Agronomic Resilience and Resource Conservation

By coupling NREL SAM with DSSAT crop models, the research will empirically validate the biological impacts of agrivoltaic shading in a semi-arid climate. The outputs will establish comprehensive yield-impact factors for staple crops (cotton and corn) relative to a baseline without agrivoltaics and shade. The anticipated data will demonstrate how optimized shading configurations can mitigate severe heat stress and substantially reduce agricultural water demand, establishing agrivoltaics as an active climate-adaptation tool for drought-prone regions.

4.3 A Multi-Objective Optimization Tool for Dual-Use Land

The culmination of Research Objective 2 will be the deployment of a functional, open-architecture Two-Stage Stochastic MILP solver. This algorithmic framework will process complex, competing variables across energy generation, agricultural yield, and human/machinery interference. The resulting tool will be able to generate Pareto-optimal system layouts, mathematically demonstrating that when agricultural preservation and water conservation are

properly valued, broad-acre agrivoltaics represent a highly profitable and resilient investment strategy.

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